

Math 196 Optional Problem Set 9

1 Review of Earlier Topics

Problem 1. Solve the following linear systems:

(a)
$$\begin{cases} 2x_1 + 2x_2 + 2x_3 - x_5 = 4 \\ x_4 - x_5 = -1 \end{cases}$$

(b)
$$\begin{bmatrix} 3 & 0 & -1 \\ 1 & 1 & 1 \\ -2 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Problem 2. Consider the matrices

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 3 & 0 & 1 \end{bmatrix}, S = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Compute $S^{-1}AS$.

Problem 3. Find a vector $v \in \mathbb{R}^2$ such that if you first apply a reflection across the line

$$\ell = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

and then apply a rotation through angle $\pi/3$, v is sent to the vector $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Problem 4. Find bases for the image and kernel of the following matrix:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 1 & -1 & 0 & 2 \\ 0 & 2 & 2 & -4 \end{bmatrix}$$

Problem 5. Consider the transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y - z \\ z - x \\ x - y \end{bmatrix}$$

1. Find the matrix representing T with respect to the standard basis.
2. Find the matrix representing T with respect to the basis

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Problem 6. Find an orthonormal basis for

$$W = \ker \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}$$

and use this to compute the orthogonal projection of

$$\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

onto W .

Problem 7. Find a least squares approximate solution to the linear system

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Problem 8. Find the line $y = ax + b$ which best fits the points

$$(-1, 1), (0, 1), (1, 2)$$

using the method of least squares.

Problem 9. For which values of t is the matrix

$$\begin{bmatrix} 1 & 1 & 1+t \\ t & -1 & 0 \\ 3 & -1-t & 1 \end{bmatrix}$$

invertible?

Problem 10. Compute the determinant of the following matrix using any method.

$$\begin{bmatrix} 2 & -2 & 4 & 4 \\ 1 & 1 & -1 & 2 \\ 1 & 0 & 1 & -1 \\ 2 & -1 & 3 & 4 \end{bmatrix}$$

Problem 11. Suppose that $\tilde{x}$ is a least squares approximate solution to the linear system $Ax = v$, and suppose that O is an orthogonal matrix. Prove that $\tilde{x}$ is also a least squares approximate solution to $OAx = Ov$.

Problem 12. In each part, either find an example of the thing it describes or explain why it is impossible.

(a) An invertible matrix S such that

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = S \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} S^{-1}$$

(b) An orthogonal matrix A such that

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$
$$A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

(c) A 2×4 matrix whose kernel is spanned by the vector

$$\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

(d) Two orthogonal matrices A and B such that $AB \neq BA$.

(e) A matrix A such that $A^2 + 2A$ is the zero matrix, and 3 is an eigenvalue of A .

Problem 13. Let A and B be $n \times n$ matrices.

(a) Suppose that $AB = BA$. Let V_λ be an eigenspace for A . Show that if $v \in V_\lambda$, then $Bv \in V_\lambda$.

(b) Suppose that $AB - BA = kB$ for some number k , and that v is an eigenvector for A with eigenvalue λ . Show that Bv is also an eigenvector for A . What is the associated eigenvalue?

2 Problems on Week 9 Topics

Problem 14. Evaluate each expression. In the answer, complex numbers should be written as $a + bi$, for $a, b \in \mathbb{R}$.

(a) $(1 - i)(2 + i) + 3i$

(b) $\frac{(2i+1)(1-i)+2i}{3-i}$

(c) $\begin{bmatrix} 2 & 1-i \\ i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} -i \\ 1+2i \end{bmatrix}$

Problem 15. Describe the set of solutions to each linear system, where x, y are chosen from $\mathbb{C}$.

(a) $\begin{bmatrix} 1 & 1 \\ -i & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1+i \\ -1+i \end{bmatrix}$

(b) $\begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3+i \\ 3i-1 \end{bmatrix}$

Problem 16. For each matrix, find all real and complex eigenvalues and diagonalize if possible.

(a) $\begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & -4 & 3 \\ 2 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Problem 17. Let R be the rotation¹ $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ about the x -axis through angle $\pi/3$, and let S be the rotation about the z -axis through angle $\pi/2$. Compute the product SR . This is a rotation; find the axis of rotation and angle. (The angle can be written as $\cos^{-1}$ of something.)

Problem 18. The Lucas numbers are defined by the same recurrence relation as the Fibonacci numbers, $L_n = L_{n-1} + L_{n-2}$, with a different initial condition: $L_0 = 2$, $L_1 = 1$. Find an explicit formula for L_n in terms of n .

Problem 19. A population of birds is foraging on a group of three islands. Each morning, all of the birds on each island fly to a different island, dividing themselves evenly between the other two islands.

- (a) Let $x(n)$, $y(n)$, and $z(n)$ represent the number of birds on each island on day n . Find a transition matrix that expresses $x(n+1)$, $y(n+1)$, and $z(n+1)$ as a function of $x(n)$, $y(n)$ and $z(n)$.
- (b) Find the eigenvectors and eigenvalues of the transition matrix.
- (c) If on the first day, the whole population of 900 birds is on one of the islands, explain how the birds will be distributed across the three islands after many days have passed.

¹see Problem Set 2, Problem 6 for the formulas for the rotation matrices